

Unit 1 Guided Notes — The Real Number System

Classify it, root it, estimate it, convert it, and order it.

Standards: NC.8.NS.1 · NC.8.NS.2

Name: _____

Date: _____

Period: _____

🎯 By the end of this unit, I can...

- Sort numbers into the real number system (rational vs. irrational)
- Find perfect squares, perfect cubes, and their roots
- Estimate square roots on a number line and to the nearest tenth
- Convert between fractions and decimals, including repeating decimals
- Compare and order real numbers from least to greatest

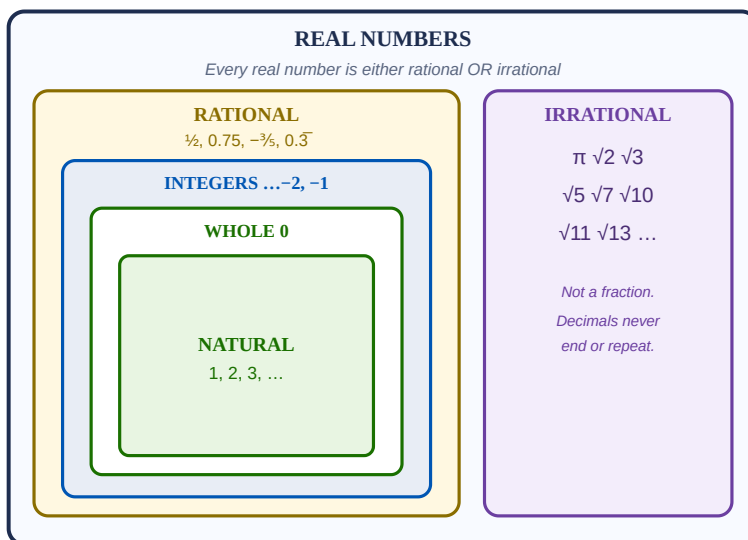
Fill in the blanks and show your work in the space provided. Worked examples are done for you — use them as models.

1 Number Classification

Every real number belongs to one or more of these sets. Write the definition for each:

Set	Examples	Definition
Natural Numbers	1, 2, 3, 4, ...	
Whole Numbers	0, 1, 2, 3, ...	
Integers	..., -2, -1, 0, 1, 2, ...	
Rational Numbers	$\frac{1}{2}$, 0.75, $-\frac{3}{5}$, $0.\overline{3}$	
Irrational Numbers	π , $\sqrt{2}$, $\sqrt{3}$	

The Real Number System — Nested Sets



Match the Set to its Definition

Draw a line from each set to the matching definition.

Set

Natural Numbers

Whole Numbers

Integers

Irrational Numbers

Definition

Whole numbers + their negatives

Counting numbers: 1, 2, 3, ...

Non-repeating, non-terminating decimal

Naturals + zero

CHECK Which number is *irrational*? Circle one.

0.55

$\sqrt{49}$

$\sqrt{7}$

$-3/4$

ERROR ANALYSIS A student claims: " $\sqrt{49}$ is irrational because it has a square-root symbol." What is the mistake?

2 Perfect Squares & Square Roots

 **THE RELATIONSHIP:** Squaring and square-rooting are INVERSE OPERATIONS — they undo each other. If $5^2 = 25$, then $\sqrt{25} = 5$.

A **perfect square** is a whole number times itself ($n \times n = n^2$). Its **square root** takes you back to that whole number, so $\sqrt{(\text{a perfect square})}$ is always a whole number.

Perfect Squares 1–15 — Reference Chart

n	n ²	n	n ²	n	n ²
1	1	6	36	11	121
2	4	7	49	12	144
3	9	8	64	13	169
4	16	9	81	14	196
5	25	10	100	15	225

 **READ IT BOTH WAYS:** $12^2 = 144$ means $\sqrt{144} = 12$. Every row is two facts in one.

CHECK Circle the perfect square:

$\sqrt{169} =$ _____

30

50

144

200

👉 **NOT A PERFECT SQUARE?** Then the root is irrational (like $\sqrt{7}$ or $\sqrt{20}$) — we estimate those in Section 4.

3 Perfect Cubes & Cube Roots

📌 **THE RELATIONSHIP:** Cubing and cube-rooting are INVERSE OPERATIONS too. If $5^3 = 125$, then $\sqrt[3]{125} = 5$. The cube-root symbol carries a small 3: $\sqrt[3]{}$.

A **perfect cube** is a whole number used as a factor three times ($n \times n \times n = n^3$). Its **cube root** undoes that.

Perfect Cubes 1–10 — Reference Chart

n	n^3	n	n^3
1	1	6	216
2	8	7	343
3	27	8	512
4	64	9	729
5	125	10	1000

💡 **SQUARE ROOT vs CUBE ROOT:** $\sqrt{64}$ asks “what number times *itself* = 64?” → 8. $\sqrt[3]{64}$ asks “what number used *three times* = 64?” → 4. Same number underneath, different answer — watch the little 3!

$$\sqrt[3]{27} = \underline{\hspace{2cm}}$$

$$\sqrt[3]{1000} = \underline{\hspace{2cm}}$$

4 Estimating Square Roots

📌 **THE BIG IDEA:** When a number is NOT a perfect square, its square root is irrational — it never ends or repeats. We can't write it exactly, but we can *estimate* it. Two tools: a number line and a step-by-step algorithm.

Method 1 — Place it on a Number Line

1. Find the two perfect squares your number lands **between**.
2. Take the $\sqrt{}$ of each — those are two back-to-back whole numbers. Your root lands between them.
3. Decide which one it's **closer** to: is your number nearer the lower or the upper perfect square?
4. Place it on the number line.

Example (model): place $\sqrt{20}$

$16 < 20 < 25$, so $\sqrt{20}$ is between 4 and 5. Since 20 is closer to 16, $\sqrt{20}$ sits a little below the middle — closer to 4.



Method 2 — The Estimating Algorithm (to the nearest tenth)

Color key: **gold = perfect squares** · **green = the roots** · **blue = your number**.

Example (model): estimate $\sqrt{45}$

Step 1 — Trap your number between two perfect squares.

$$36 < 45 < 49$$

$\sqrt{36} = 6$ · $\sqrt{49} = 7 \rightarrow \sqrt{45}$ is between 6 and 7. Smaller root = 6.

Step 2 — Build the fraction: how far in is your number?

your number – lower square

$$45 - 36 = 9$$

$$49 - 36 = 13$$

upper square – lower square

$$= 9 \div 13 \approx 0.7$$

The fraction is always squares on top and bottom — never the roots.

Step 3 — Add the fraction to the smaller root.

$$\sqrt{45} \approx 6 + 0.7 = 6.7 \quad (\text{calculator: } 6.708...)$$

💡 IN SHORT: $\sqrt{n} \approx (\text{smaller root}) + (n - \text{lower square}) \div (\text{upper square} - \text{lower square})$. This is an *estimate* — it can be off by about a tenth from a calculator, and that's fine.

⚠️ DON'T MIX THEM UP: the number you **add** is a **root**. The numbers **inside the fraction** are the **perfect squares**. Roots to add, squares to subtract.

YOUR TURN Estimate $\sqrt{30}$ to the nearest tenth. (Hint: $25 < 30 < 36$) — show your steps below.

$\sqrt{30} \approx$ _____

 **NEGATIVE ROOTS — JUST FLIP THE SIGN:** estimate the positive root first, then make it negative.

Example: $-\sqrt{40} \rightarrow \sqrt{40} \approx 6.3$, so $-\sqrt{40} \approx -6.3$.

 **WHICH METHOD?** Use the number line for “between which whole numbers.” Use the algorithm for a decimal to the nearest tenth.

5 Converting Fractions $\leftrightarrow$ Decimals

Fraction $\rightarrow$ Decimal

Divide the numerator by the denominator (top $\div$ bottom).

Fraction	Division	Decimal
$3/4$	$3 \div 4$	
$1/8$	$1 \div 8$	
$1/3$	$1 \div 3$	

 **KEY INSIGHT:** Repeating decimals are _____ because they can be written as fractions. Non-repeating, non-terminating decimals (like π) are _____. Use a **bar** over the digits that repeat: $0.\overline{3}$ means $0.3333\dots$

Decimal $\rightarrow$ Fraction (terminating)

Decimal	Read / write	Fraction (simplest form)
0.75	75 hundredths $\rightarrow 75/100$	
0.6	6 tenths $\rightarrow 6/10$	
0.125	125 thousandths $\rightarrow 125/1000$	

Repeating Decimal $\rightarrow$ Fraction: The 9's Rule

 **THE 9's RULE:** the repeating digits go in the NUMERATOR. The DENOMINATOR is made of 9's — one 9 for each repeating digit.

Repeating digits	Denominator	Example	Simplified
1 digit		$0.\overline{3} = 3/9$	
2 digits		$0.\overline{27} = 27/99$	
3 digits		$0.\overline{125} = 125/999$	

Worked Example (model): convert $0.\overline{27}$ ($0.272727\dots$) to a fraction

Step 1 — Count the repeating digits: 2 digits repeat (the block “27”).

Step 2 — Denominator = that many 9's: 2 digits $\rightarrow$ 99.

Step 3 — Numerator = the repeating block: 27/99.

Step 4 — Simplify: $27/99 = 3/11$.

⚠ IMPORTANT: the 9's rule only works when the decimal starts repeating *right after* the decimal point. For $0.1666\dots$ (the 1 doesn't repeat), use the algebra method. Always simplify!

CHECK Which decimal equals $3/8$? Circle one.

0.38

0.375

0.83

0.625

6 Ordering Numbers Least to Greatest

📌 THE STRATEGY: to compare a mix of fractions, decimals, mixed numbers, and roots — turn *everything* into a decimal first, then order.

Guided Example — Order $\sqrt{5}$, $2\frac{1}{4}$, $7/3$, 2.1

Number	Decimal	How?
$\sqrt{5}$		Estimating algorithm (between 4 and 9)
$2\frac{1}{4}$		$2 + (1 \div 4)$
$7/3$		$7 \div 3$
2.1		Already a decimal

Order (least $\rightarrow$ greatest): _____ < _____ < _____ < _____

Plot each on the number line, then read them left to right:



💡 MIXED NUMBERS: turn them into a decimal — whole number + (numerator $\div$ denominator). Example: $3\frac{2}{5} = 3 + (2 \div 5) = 3.4$.

⚠ **WATCH OUT:** line up the same number of decimal places before comparing, and estimate any irrational roots *first*.

ERROR ANALYSIS To order 1.5 , $\sqrt{2}$, $7/4$, $\sqrt{3}$ a student wrote $1.5 < \sqrt{2} < \sqrt{3} < 7/4$. Is this right? Explain and fix it.

Part 2 — Practice Problems

Show all work. Circle your final answer.

A Classify Each Number

Rational = can be written as a fraction (terminating or repeating decimal). Irrational = non-repeating, non-terminating.

1. $0.333\ldots$ _____

3. $\sqrt{49}$ _____

5. $-3/8$ _____

7. $\sqrt{144}$ _____

9. $2/7$ _____

11. $-\sqrt{81}$ _____

2. π _____

4. $\sqrt{10}$ _____

6. $0.1212\ldots$ _____

8. $0.101001\ldots$ _____

10. $\sqrt{2}$ _____

12. 3.14 _____

B Estimate Each Square Root (nearest tenth)

Estimate each root to the nearest tenth. For 1–6, use the number line: write the two whole numbers your root falls between in the boxes, mark the point, then read the estimate. For 7–11, use the algorithm and show your work. A leading – just flips the sign.

1. $\sqrt{5} \approx$ _____



2. $\sqrt{10} \approx$ _____



3. $\sqrt{20} \approx$ _____



4. $\sqrt{30} \approx$ _____



5. $\sqrt{50} \approx$ _____



6. $\sqrt{8} \approx$ _____



7. $\sqrt{15} \approx$ _____

8. $\sqrt{40} \approx$ _____

9. $\sqrt{65} \approx$ _____

10. $\sqrt{90} \approx$ _____

11. $-\sqrt{50} \approx$ _____

C Order from Least to Greatest

Convert each number to a decimal, then arrange from least to greatest.

1. Order: π , $\sqrt{10}$, $10/3$, 3.5

Write each as a decimal, then order them:

Least → greatest: _____

2. Order: 1.5 , $\sqrt{2}$, $7/4$, $\sqrt{3}$

Write each as a decimal, then order them:

Least → greatest: _____

3. Order: $\sqrt{5}$, 2.5 , $11/5$, $\sqrt{7}$

Write each as a decimal, then order them:

Least → greatest: _____

4. Order: 4.2 , $\sqrt{18}$, $13/3$, $\sqrt{20}$

Write each as a decimal, then order them:

Least → greatest: _____

5. Order: -2.5 , $-\sqrt{5}$, $-8/3$

Write each as a decimal, then order them:

Least → greatest: _____

6. Order: $\sqrt{50}$, 7.1, $\frac{22}{3}$, $\sqrt{52}$

Write each as a decimal, then order them:

Least → greatest: _____

7. Order: $\frac{5}{6}$, $\sqrt{0.5}$, 0.9, $\frac{7}{8}$

Write each as a decimal, then order them:

Least → greatest: _____

8. Order: $\sqrt{30}$, 5.5, $\frac{17}{3}$, $\sqrt{32}$

Write each as a decimal, then order them:

Least → greatest: _____